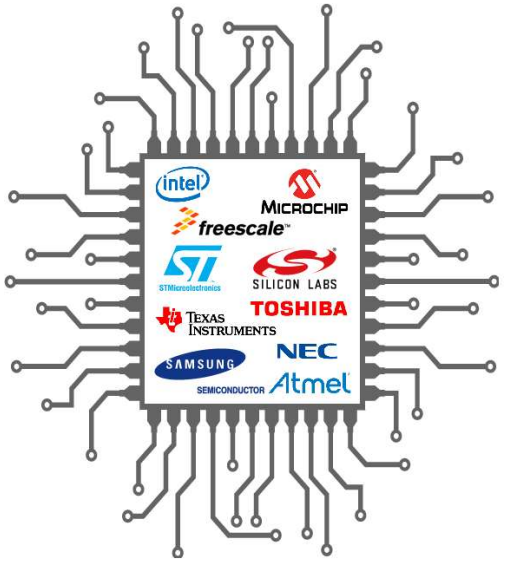
Transducers



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Learning Outcomes

- On completion of this session, you should be able to
  - Explain the role of transducers
  - Explain how different sensors capture the state of physical world
  - Explain how different actuators change the state of physical world
  - Understand the operation of commonly used sensors and actuators
  - Interface commonly used sensors and actuators with microcontrollers

Transducer

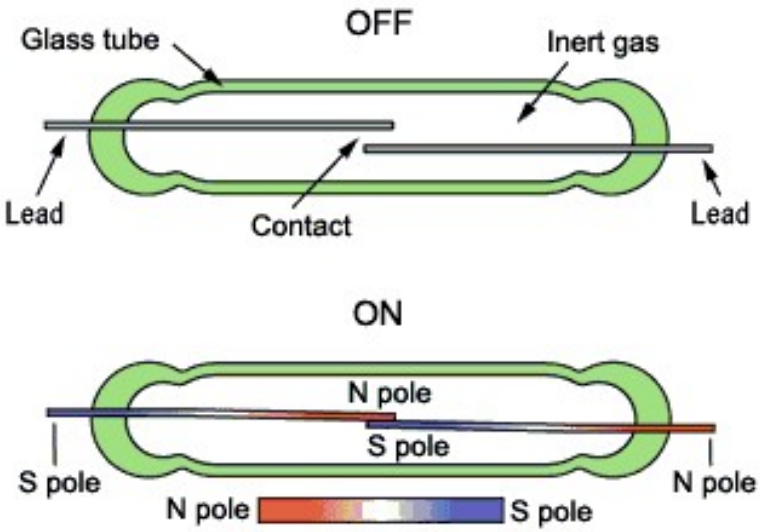
- Converts energy or physical quantities from one form to another
- Used at the boundary of an embedded system
- Categorized according to the direction of conversion
  - Sensors
  - Actuators

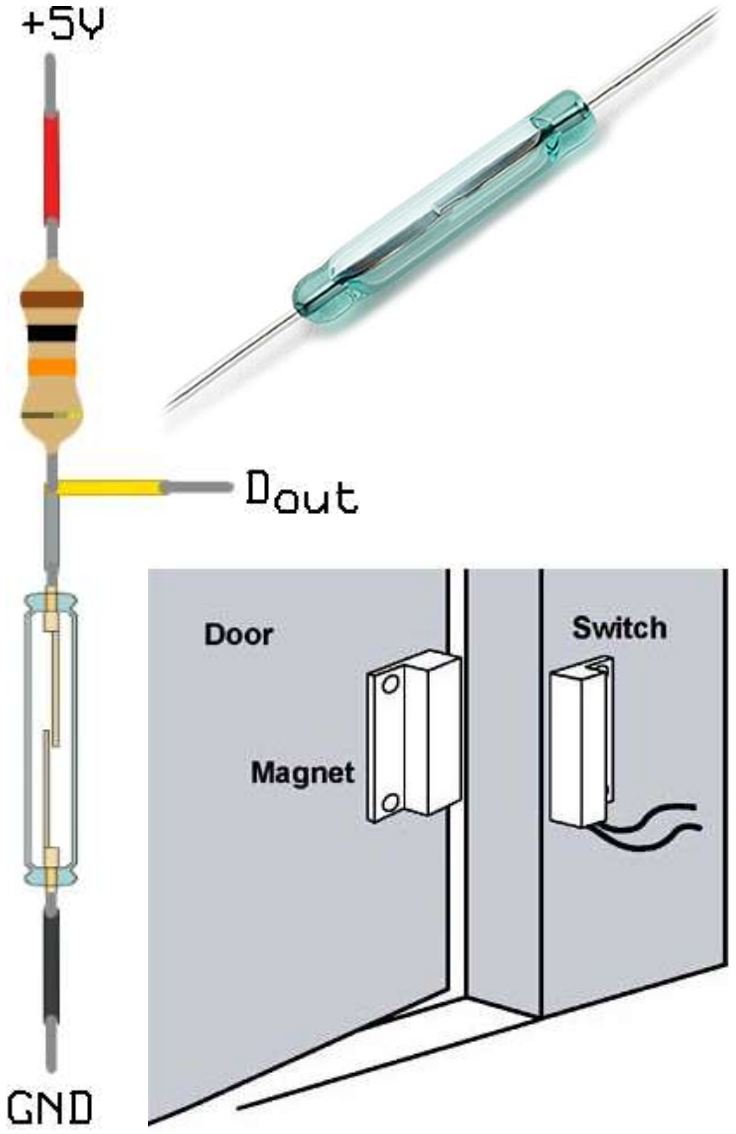
Sensors

- Convert physical quantities into electrical signals
- Used to capture state of physical world as inputs
- Two kinds
  - Digital
  - Analog

Digital Sensors

- Reed Switch
  - Normally open but closes when a magnetic field is present
  - Detects presence or absence of magnetic energy and produces two discrete voltages

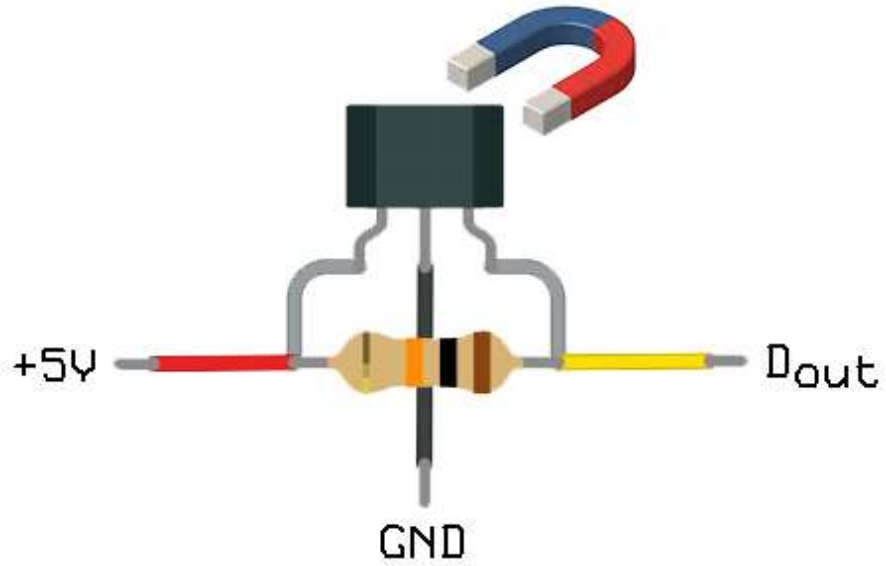




Digital Sensors

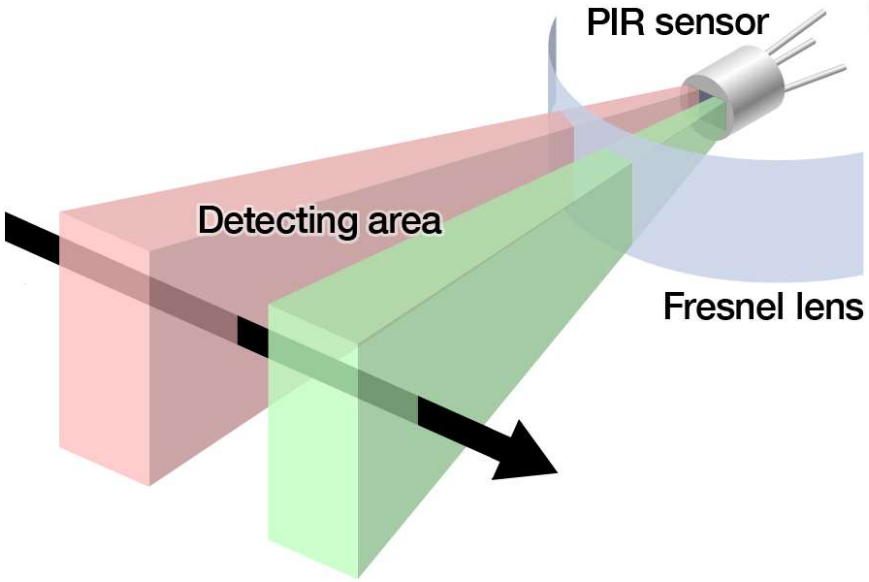
- Hall Effect Sensor
  - Detects presence or absence of magnetic energy and produces two discrete voltages





Digital Sensors

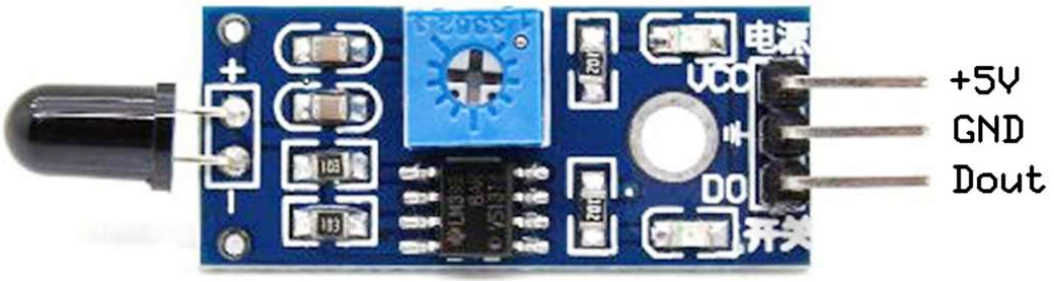
- PIR Sensor
  - Detects variation of IR radiation energy in sensor's field of view and produces two discrete voltages





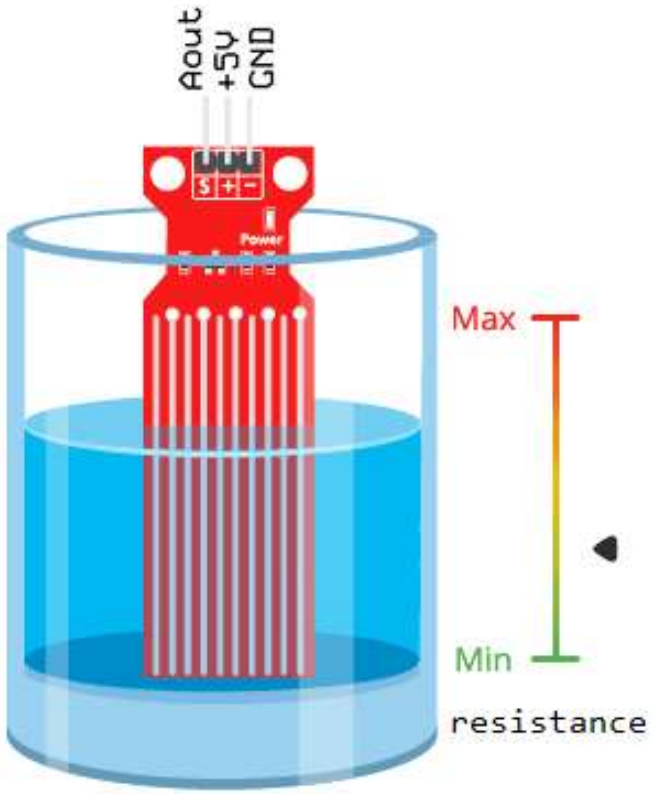
Digital Sensors

- Flame Sensor
  - Detects IR radiation energy emitted from a fire and produces two discrete voltages



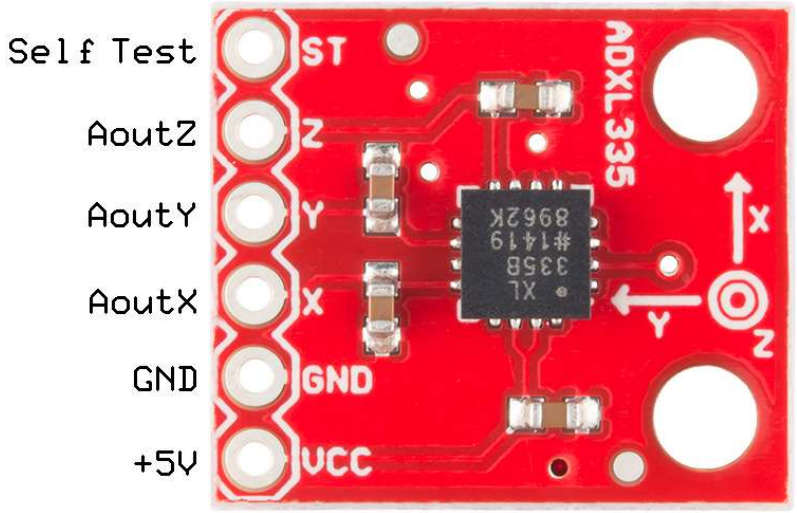
Analog Sensors

- Water Level Sensor
  - Converts continuous variation of water level into a continuous variation of voltage



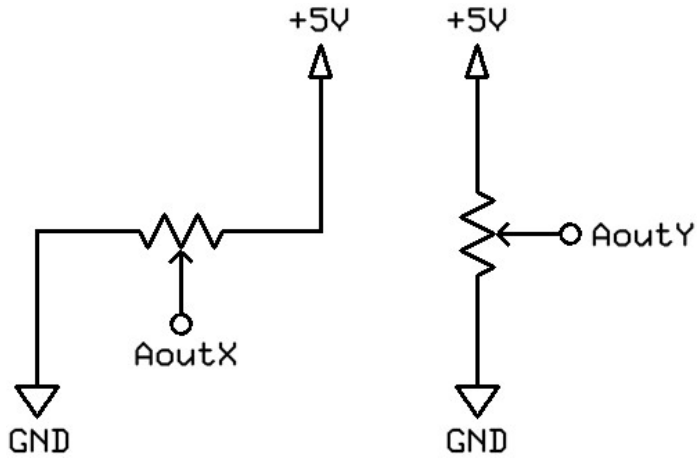
Analog Sensors

- Accelerometer
  - Converts continuous variation of linear acceleration in the 3D space into continuous variation of three voltages



Analog Sensors

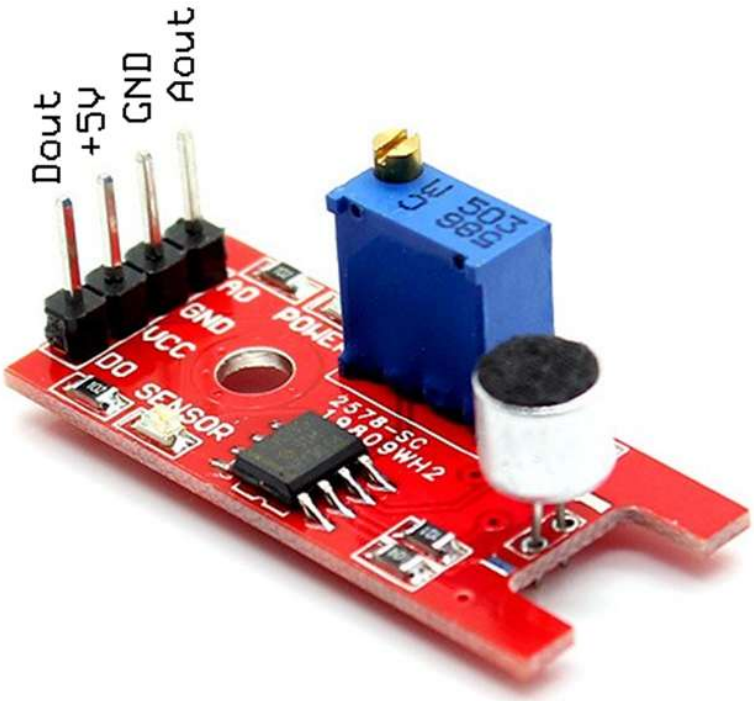
- Thumb Joystick
  - Converts continuous displacement along X and Y axes into continuous variation of two voltages





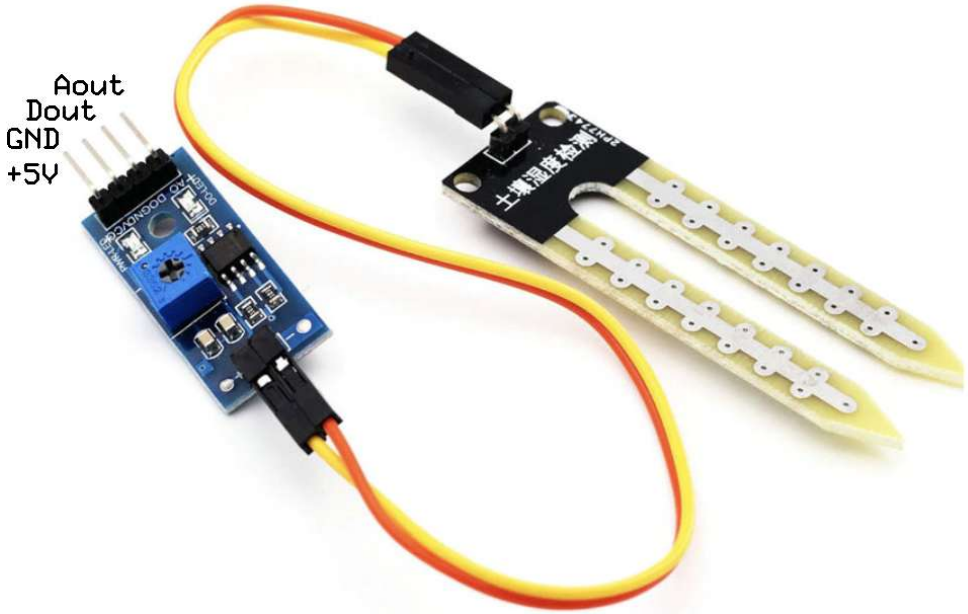
Analog/Digital Sensors

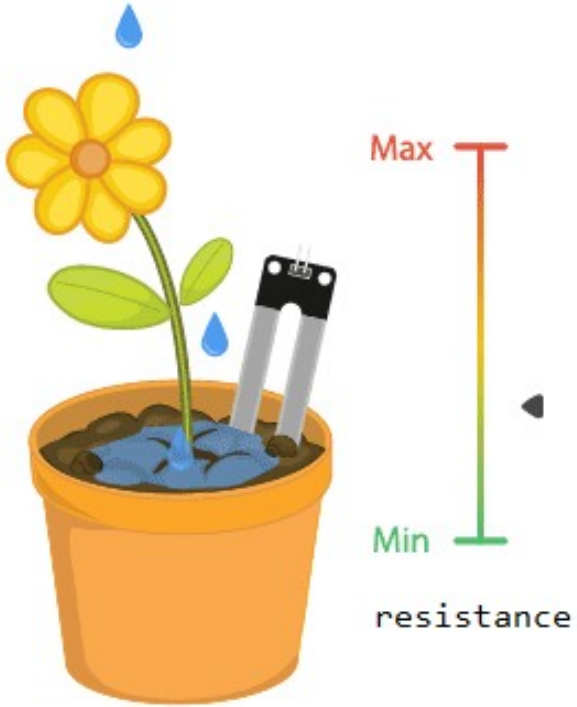
- Sound Sensor
  - Converts variation sound into a continuous variation of voltage as well as into two discrete voltages



Analog/Digital Sensors

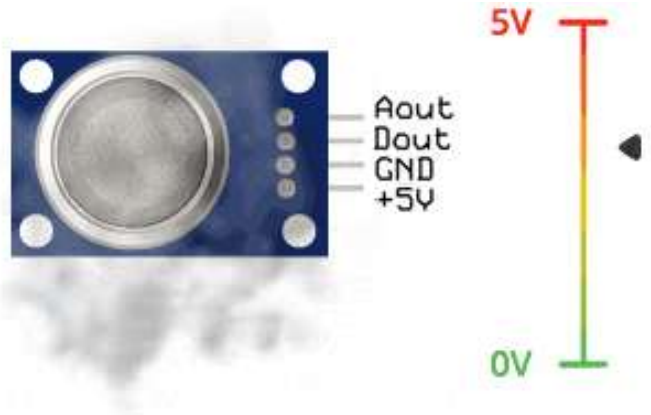
- Soil Moisture Sensor
  - Converts continuous variation of soil moisture level into a continuous variation of voltage as well as into two discrete voltages





Analog/Digital Sensors

- Gas/Smoke Sensor
  - Converts continuous variation of gas/smoke concentration into a continuous variation of voltage as well as into two discrete voltages

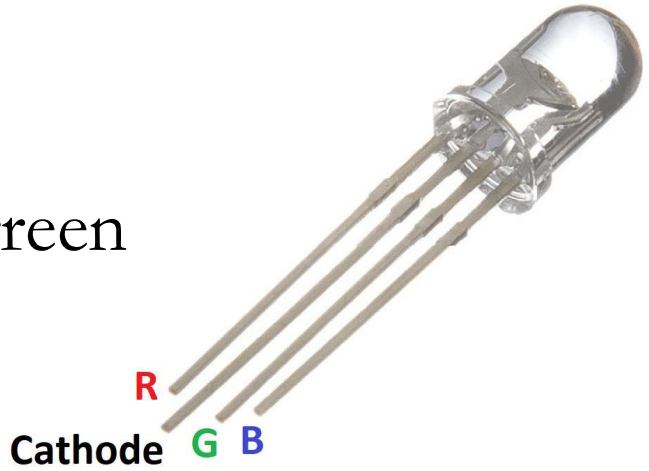


Actuators

- Convert electrical energy into other forms
- Used to change the state of physical world
- Some actuators require higher voltages and draw higher currents than an I/O port pin can deliver

Actuators

- RGB LED
  - Converts electrical energy into red, green and blue colour visible light
- Incandescent Bulb
  - Converts electrical energy into visible light and heat
  - Typically requires a higher voltage and a current than an I/O port pin can deliver





Actuators

- Motor
  - Converts electrical energy into kinetic energy (circular motion)
  - Even small ones draw more current than an I/O port pin can deliver

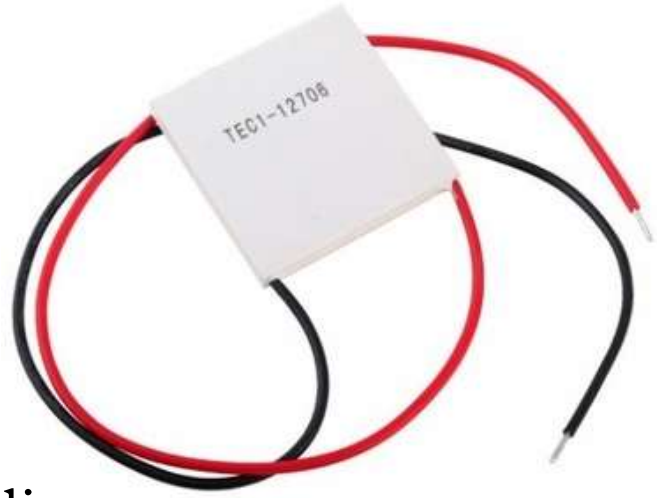


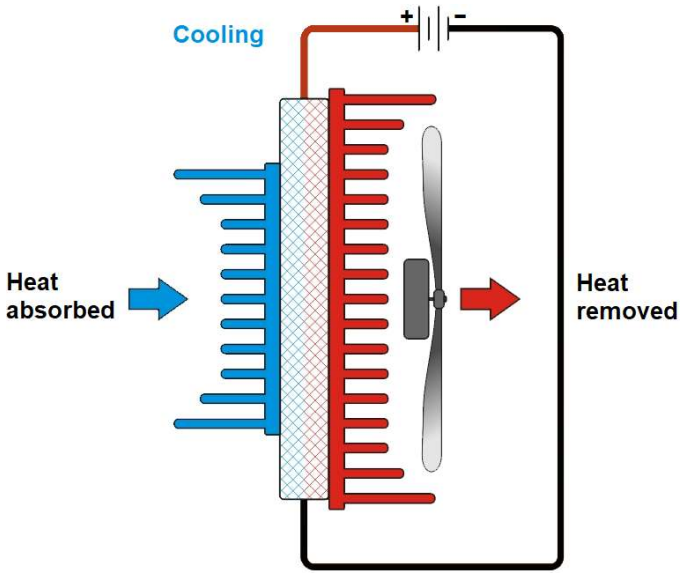
- Solenoid
  - Converts electrical energy into kinetic energy (linear motion)
  - Requires a higher voltage and a current than an I/O port pin can deliver

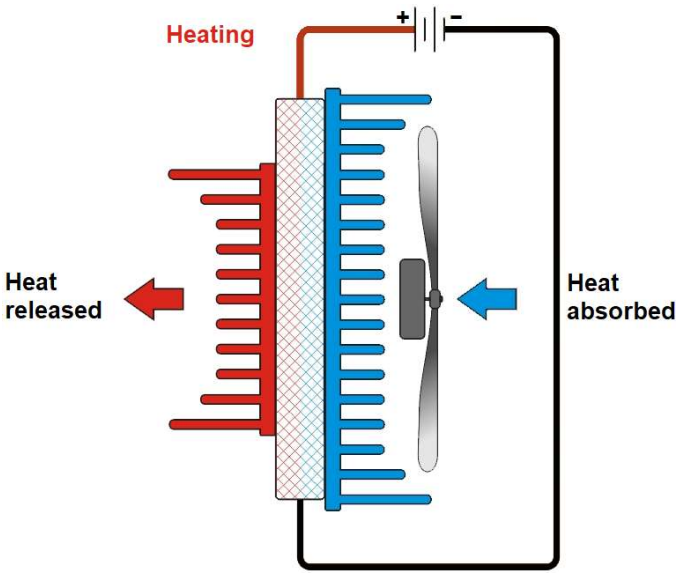


Actuators

- Thermoelectric Cooler/Heater
  - Converts electrical energy into thermal energy
  - Requires a higher voltage and a current than an I/O port pin can deliver

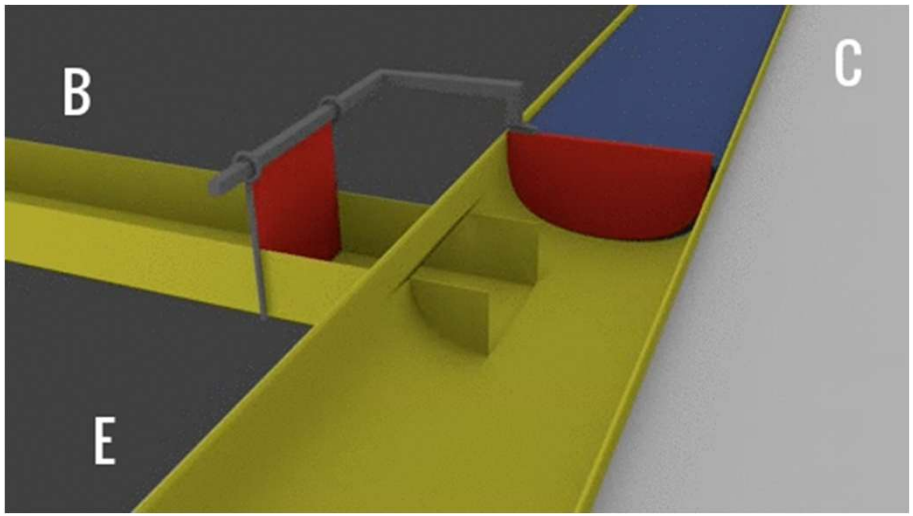


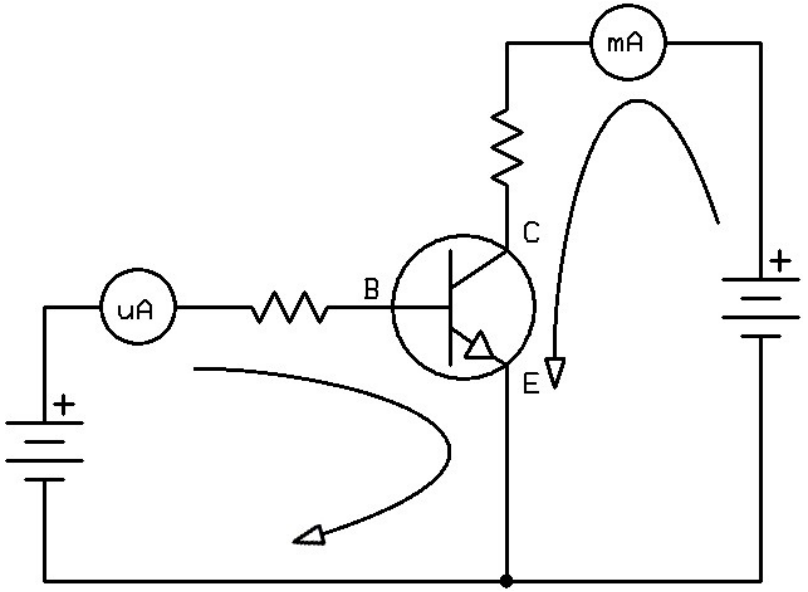


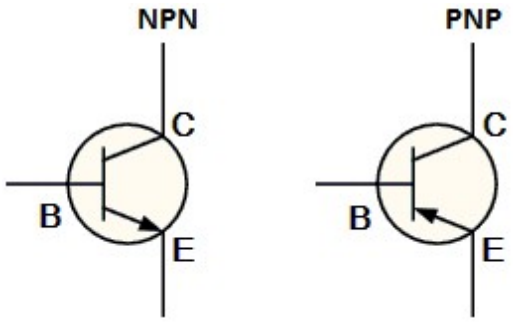


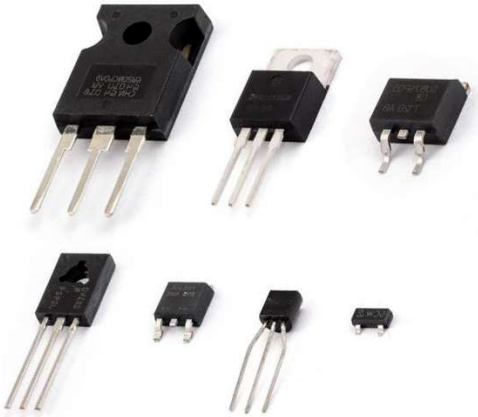
Actuators

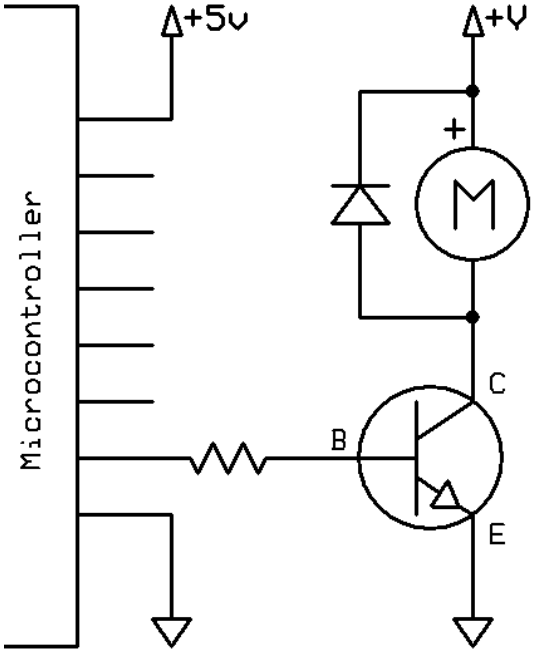
- Driving with a BJT





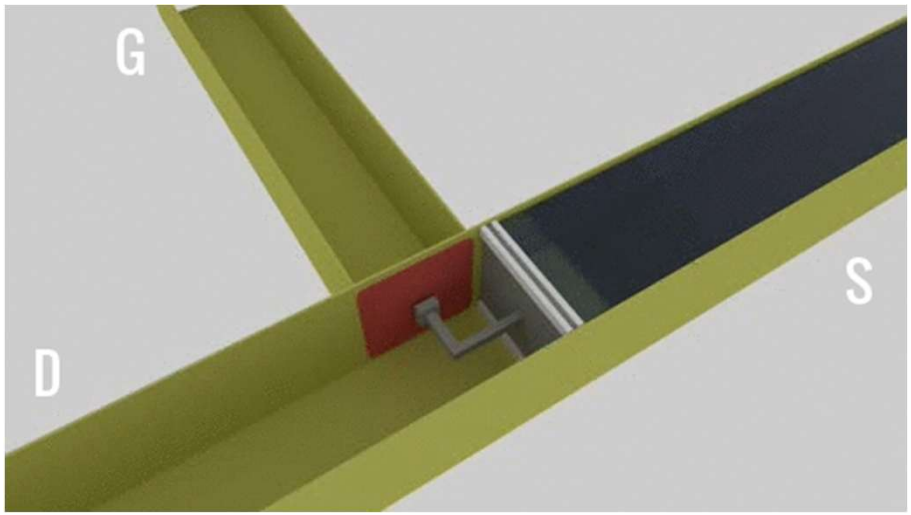


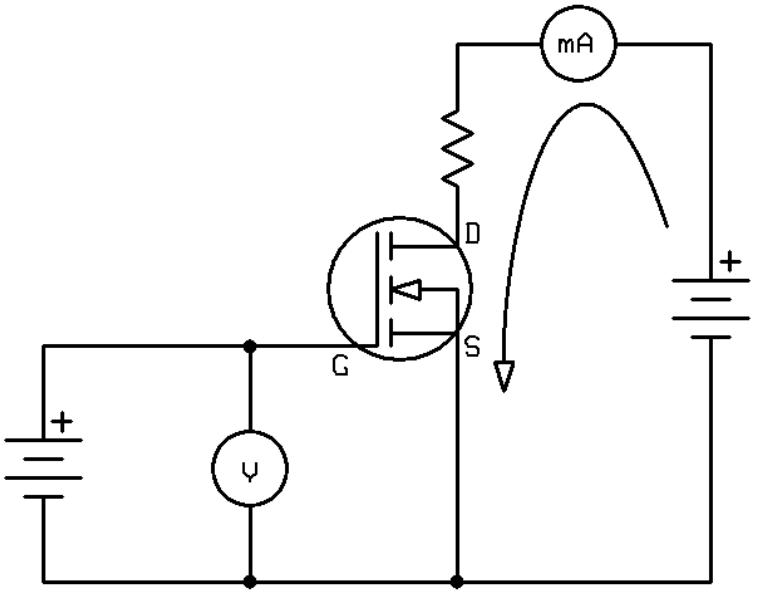


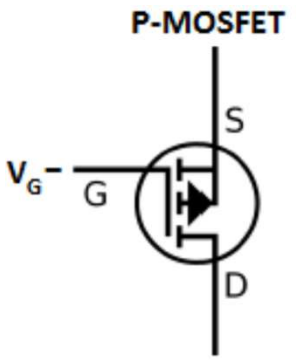


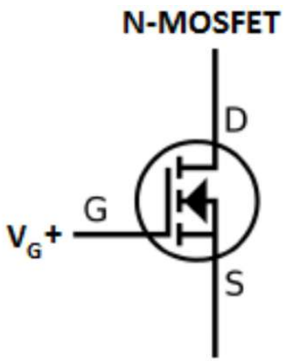
Actuators

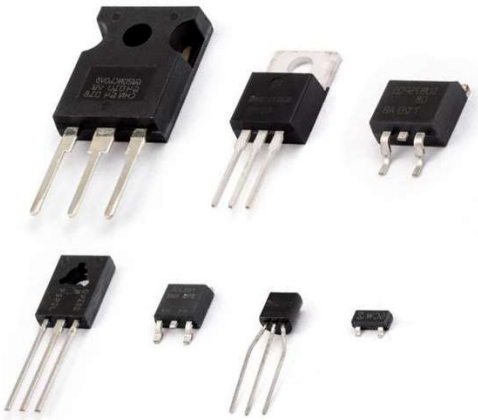
- Driving with a MOSFET

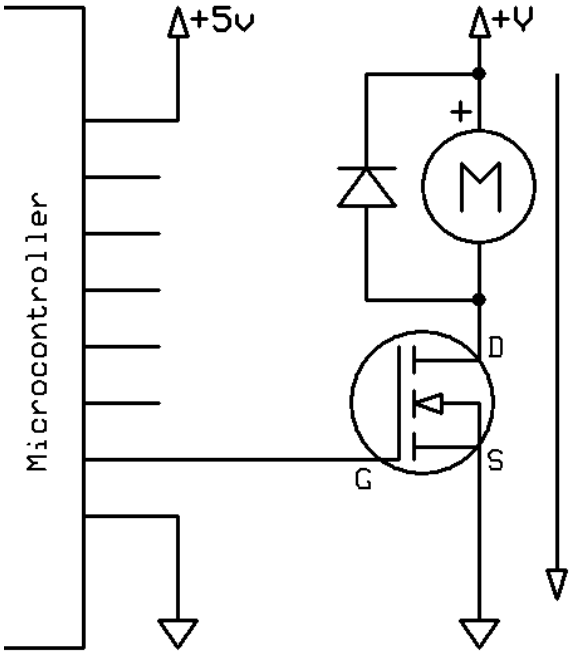






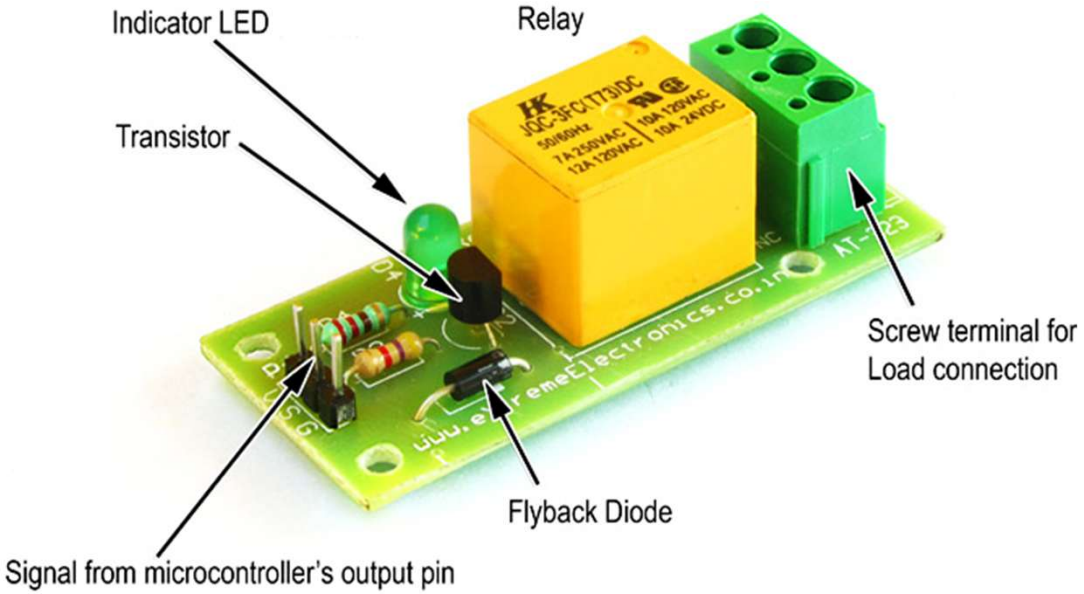


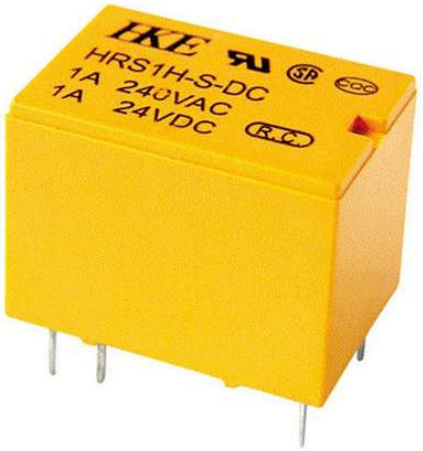


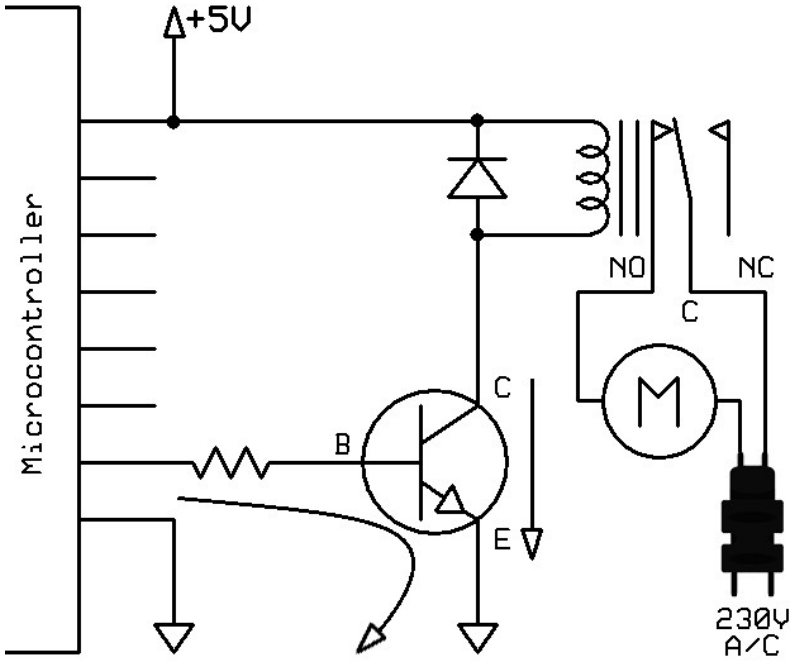


Actuators

- Switching with a Relay







Questions